

Unit 26: The Definite Integral

1. The idea, in plain words

You've now built, from scratch, the machinery for computing exact areas using limits of Riemann sums. This unit takes a step back from the raw computation and asks: once we know $\int_a^b f(x) dx$ represents a signed area, what handy rules follow just from that idea — without needing to grind through a limit every time?

A crucial nuance: the definite integral gives signed, or net, area. If the curve dips below the x-axis, that portion counts as negative area — it doesn't just vanish, and it doesn't add like ordinary area would. Picture a hiker's elevation change: walking uphill counts positive, walking downhill counts negative, and the "total" is the net change, not the total distance covered.

This unit collects a toolkit of properties that let you combine, split, flip, and bound integrals — all without recomputing anything from the limit definition. Two of these are especially powerful:

The Max-Min Inequality (bounding an integral without evaluating it): if you know the smallest value (m) and largest value (M) that f takes on $[a, b]$, then the exact area has to be squeezed between two easy-to-compute rectangles — one using the shortest possible height, one using the tallest. Picture the true curve trapped between a short flat rectangle underneath it and a tall flat rectangle above it.

The odd/even symmetry shortcut: if a function is odd (its graph has 180° rotational symmetry through the origin) and you integrate it over an interval that's perfectly symmetric around 0 (like $[-a, a]$), the positive area on one side exactly cancels the negative area on the other — the whole integral collapses to 0, with zero computation required. If the function is even instead (mirror-symmetric across the y-axis), you can cut your work in half by just computing one side and doubling it.

2. Toolbox

Definite integral as signed area: $\int_a^b f(x) dx$ is positive where f is above the x-axis, negative where f is below it.

Core properties:

$$\int_a^a f(x) dx = 0$$

$$\int_a^b f(x) dx = -\int_b^a f(x) dx \quad (\text{reversing the limits flips the sign})$$

$$\int_a^b c \cdot f(x) dx = c \int_a^b f(x) dx$$

$$\int_a^b [f(x) \pm g(x)] dx = \int_a^b f(x) dx \pm \int_a^b g(x) dx$$

$$\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx \quad (\text{splitting/joining over adjacent intervals})$$

Max-Min Inequality: if $m \leq f(x) \leq M$ for every x in $[a, b]$, then

$$m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$$

Odd/Even symmetry (only valid on a symmetric interval $[-a, a]$):

$$f \text{ odd } (f(-x) = -f(x)) \implies \int_{-a}^a f(x) dx = 0$$

$$f \text{ even } (f(-x) = f(x)) \implies \int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

Classifying combinations of odd/even functions (very useful for the symmetry shortcut):

$$\text{odd} \times \text{odd} = \text{even} \quad \text{even} \times \text{even} = \text{even} \quad \text{odd} \times \text{even} = \text{odd}$$

$$\text{odd} + \text{odd} = \text{odd} \quad \text{even} + \text{even} = \text{even}$$

Note: x^n is even when n is even, and odd when n is odd.

3. Common mistakes

- Forgetting the sign flip when reversing the limits of integration. $\int_b^a f(x) dx = -\int_a^b f(x) dx$, not the same value.
- Treating the integral as ordinary (unsigned) area. Remember: area below the x-axis subtracts from the total, it doesn't add.
- Using the wrong m and M for the Max-Min Inequality — these must be the actual minimum and maximum of f on the given interval, which usually requires checking whether f is increasing, decreasing, or has a peak/valley inside the interval — not just guessing from the endpoints blindly.
- Misclassifying a function as odd or even, especially for products or sums of several pieces. Test methodically: check each piece separately, then combine using the odd/even combination rules.
- Applying the odd/even shortcut on a non-symmetric interval. This trick only works when the interval is exactly $[-a, a]$, centered at 0 — it does not apply to something like $[-1, 3]$.

4. Problem Set

Warm-up

1. Given $\int_1^4 f(x) dx = 10$, find $\int_4^1 f(x) dx$.
2. Given $\int_0^3 f(x) dx = 7$, find $\int_0^3 5f(x) dx$.
3. Evaluate $\int_2^2 f(x) dx$.
4. Given $\int_0^2 f(x) dx = 4$ and $\int_2^5 f(x) dx = 9$, find $\int_0^5 f(x) dx$.

5. Given $\int_1^3 f(x) dx = 6$ and $\int_1^3 g(x) dx = 2$, find $\int_1^3 [f(x) - g(x)] dx$.
6. Given $\int_0^4 f(x) dx = 8$ and $\int_0^4 g(x) dx = 3$, find $\int_0^4 [2f(x) + 3g(x)] dx$.

Standard

7. Given $\int_0^5 f(x) dx = 12$ and $\int_3^5 f(x) dx = 4$, find $\int_0^3 f(x) dx$.
8. Given $\int_1^6 f(x) dx = 20$ and $\int_1^4 f(x) dx = 9$, find $\int_4^6 f(x) dx$.
9. Given $\int_2^7 f(x) dx = 15$, find $\int_7^2 f(x) dx$ and $\int_7^2 [-3f(x)] dx$.
10. If $g(x) = f(x) + 2$, explain (using the properties) how $\int_1^5 g(x) dx$ compares to $\int_1^5 f(x) dx$, and find the exact difference between them.
11. Suppose f is continuous, $f(x) \geq 0$ for all x in $[-3, 3]$, and $\int_{-3}^3 f(x) dx = 0$. What can you conclude about f on this interval? Explain your reasoning.
12. Given that $m = 2$ and $M = 6$ are the minimum and maximum values of $f(x)$ on $[1, 4]$, use the Max-Min Inequality to find the best bounds you can state for $\int_1^4 f(x) dx$.

Challenge

13. Show that $1 \leq \int_0^1 \sqrt{x^2 + 1} dx \leq \sqrt{2}$.
14. Use the Max-Min Inequality to find bounds for $\int_0^2 (x^2 + 3) dx$.
15. Determine, without evaluating the integral directly, whether $\int_{-2}^2 (x^3 - 4x) dx = 0$. Justify your answer.
16. Determine whether $\int_{-\pi}^{\pi} x^2 \cos x dx$ can be simplified using symmetry, and explain what the simplification tells you (note: this integral does not vanish to zero — explain why not, and what it simplifies to instead).
17. Explain why $\int_{-1}^1 \left(x^{2024} \sin x + \frac{x^7}{1+x^2} \right) dx = 0$, using odd/even classification for each term — without attempting to actually evaluate any antiderivative.
18. Suppose $2 \leq f(x) \leq 5$ for all x in $[0, 6]$, and you're told $\int_0^6 f(x) dx = 20$. Is this consistent with the Max-Min Inequality? Show your reasoning.

Applied

19. A company's daily profit rate $p(t)$ satisfies $-50 \leq p(t) \leq 200$ for all t in a 30-day month. Use the Max-Min Inequality to find bounds on the total monthly profit, $\int_0^{30} p(t) dt$.

20. A temperature deviation function $d(t)$ (measuring how far the temperature is from the daily average, with $t = 0$ representing noon) is modeled as an odd function on $[-6, 6]$ (representing 6 AM to 6 PM), based on the assumption that mornings run as far below average as evenings run above it. Find $\int_{-6}^6 d(t) dt$ without further calculation, and interpret what this result means physically.
21. Pump A delivers water at a rate with $\int_0^{10} r(t) dt = 150$ gallons over 10 minutes, and Pump B delivers water with $\int_0^{10} s(t) dt = 90$ gallons over the same period. Find the total combined output, $\int_0^{10} [r(t) + s(t)] dt$.
22. A vehicle's velocity satisfies $40 \leq v(t) \leq 65$ mph for all t during a 3-hour trip. Use the Max-Min Inequality to find bounds on the total distance traveled, $\int_0^3 v(t) dt$.